

UI/UX Design Plan

Hospital Bed Allocation System

Version: 1.0

Date: 2026-05-17

Target Platform: Java Swing Desktop Application

Design Philosophy: Modern Flat UI with Medical-Grade Clarity

1. Design System

1.1 Color Palette

Token	Hex	Usage
Primary	#2563EB	Buttons, active states, headers, action items
Primary Hover	#1D4ED8	Button hover states
Success	#10B981	Available beds, success messages, discharge confirmations
Danger	#EF4444	Occupied beds, errors, critical alerts
Warning	#F59E0B	Maintenance beds, waitlist items, cautions
Neutral-900	#111827	Primary text, headings
Neutral-600	#4B5563	Secondary text, labels
Neutral-400	#9CA3AF	Disabled states, placeholders
Neutral-200	#E5E7EB	Borders, dividers
Neutral-100	#F3F4F6	Table alternating rows, card backgrounds
Neutral-50	#F9FAFB	Main window background
Surface	#FFFFFF	Panels, cards, dialogs
Overlay	rgba(0,0,0,0.5)	Modal backdrops

1.2 Typography

Element	Font	Size	Weight	Color
Window Title	Segoe UI / System	20px	Bold	Neutral-900
Section Header	Segoe UI	16px	Semibold	Neutral-900
Card Title	Segoe UI	14px	Semibold	Neutral-900
Body Text	Segoe UI	13px	Regular	Neutral-600
Table Header	Segoe UI	12px	Semibold	Neutral-600
Table Cell	Segoe UI	13px	Regular	Neutral-900
Button Text	Segoe UI	13px	Semibold	White / Primary
Label	Segoe UI	12px	Medium	Neutral-600

Table 2 – continued

Element	Font	Size	Weight	Color
Badge Text	Segoe UI	11px	Semibold	White

1.3 Spacing System

Token	Value	Usage
xs	4px	Icon padding, tight gaps
sm	8px	Inner element padding
md	16px	Card padding, form field gaps
lg	24px	Section gaps, dialog padding
xl	32px	Major section separators
2xl	48px	Page-level padding

1.4 Component Styles

Primary Button

- Background: #2563EB, Text: White, 13px Semibold
- Padding: 10px vertical, 20px horizontal
- Border radius: 6px
- Hover: Background #1D4ED8, slight shadow 0 2px 4px rgba(37,99,235,0.3)
- Pressed: Background #1E40AF

Secondary Button

- Background: White, Border: 1px #E5E7EB, Text: #4B5563
- Hover: Background #F9FAFB, Border #D1D5DB

Danger Button

- Background: #EF4444, Text: White
- Hover: #DC2626

Status Badge

- Padding: 4px 10px, Border radius: 12px (pill shape)
- Font: 11px Semibold
- Available: Background #D1FAE5, Text #065F46
- Occupied: Background #FEE2E2, Text #991B1B
- Maintenance: Background #FEF3C7, Text #92400E

Input Field

- Background: White, Border: 1px #E5E7EB
- Border radius: 6px, Padding: 10px 14px
- Focus: Border #2563EB, subtle blue glow 0 0 0 3px rgba(37,99,235,0.1)
- Error: Border #EF4444, glow red

Card Panel

- Background: White, Border: 1px #E5E7EB
- Border radius: 8px
- Shadow: 0 1px 3px rgba(0,0,0,0.1)
- Padding: 20px

Data Table

- Header: Background #F9FAFB, Text #4B5563, bottom border #E5E7EB
- Row height: 44px
- Alternating rows: White / #F9FAFB
- Hover row: #EFF6FF (very light blue)
- Selected row: #DBEAFE with left border 3px #2563EB

2. Screen Inventory

2.1 Screen List

#	Screen Name	Purpose	Priority
1	Login Screen	Authenticate user (admin/receptionist)	High
2	Dashboard	Real-time bed overview, stats, quick actions	High
3	Admit Patient Dialog	Register and admit a new or existing patient	High
4	Discharge Dialog	Process patient discharge with options	High
5	Patient Manager	Search, view, edit patient records	Medium
6	Waitlist Panel	View and manage waiting patients	High
7	Reports Center	Generate and view occupancy/turnover reports	Medium
8	Admin Panel	Manage wards, beds, staff, maintenance	Medium
9	Bed Detail View	Detailed view of a single bed' s history	Low

3. Screen Specifications

3.1 Login Screen

Layout: Centered card on neutral background

Window Size: 400×500, Non-resizable, Centered on screen

Background: #F3F4F6 (full window)

Elements (Top to Bottom):

1. **App Icon** — 64×64 hospital cross icon, centered, color #2563EB
2. **Title** — “Hospital Bed Manager”, 20px Bold, #111827, centered
3. **Subtitle** — “Sign in to your account”, 13px Regular, #4B5563, centered, margin-top 8px
4. **Card Container** — White, rounded 12px, shadow, padding 32px, width 340px centered
 - **Role Label** — “Select Role”, 12px Medium, #4B5563
 - **Role Toggle** — Two buttons side by side: “Receptionist” | “Admin”. Selected state: Primary background, white text. Unselected: #F3F4F6 background, #4B5563 text. Border radius 6px.
 - **Username Label** — “Username”, 12px Medium
 - **Username Input** — Standard input field, placeholder “Enter username”
 - **Password Label** — “Password”, 12px Medium
 - **Password Input** — Standard input field, placeholder “Enter password”, masked characters
 - **Sign In Button** — Full width, Primary button style, text “Sign In”, margin-top 24px
 - **Error Message** — Hidden by default. Red text #EF4444, 12px, appears below button on failed login.

Interactions:

- Enter key submits form
- Failed login shakes the card horizontally (animation) and shows error
- Successful login transitions to Dashboard with fade effect

3.2 Dashboard (Main Window)

Layout: Sidebar + Main Content Area

Window Size: 1200×800, Resizable, Maximized by default

Background: #F9FAFB

3.2.1 Top Navigation Bar

- **Height:** 56px
- **Background:** White, bottom border 1px #E5E7EB
- **Left:** App logo (24px) + “Bed Manager” (14px Semibold)
- **Right:** Current user name + Role badge (pill badge, primary color) + Logout icon button

3.2.2 Left Sidebar

- **Width:** 220px
- **Background:** White, right border 1px #E5E7EB
- **Items (vertical list, 44px height each):**
 - Dashboard (icon: grid) — Active state: left border 3px #2563EB, background #EFF6FF, text #2563EB

- Admit Patient (icon: user-plus)
- Discharge (icon: user-minus)
- Waitlist (icon: clock) — Shows count badge if >0 patients waiting
- Patients (icon: users)
- Reports (icon: bar-chart)
- Admin (icon: settings) — Only visible for Admin role
- **Hover:** Background #F9FAFB
- **Icon:** 20px, #9CA3AF (inactive), #2563EB (active)
- **Text:** 13px Regular, #4B5563 (inactive), #2563EB (active)

3.2.3 Main Content Area — Dashboard View Stats Row (4 cards, horizontal, gap 20px):

Card	Icon Color	Value	Label
Total Beds	#6B7280	48	Total Beds
Available	#10B981	12	Available Now
Occupied	#EF4444	32	Occupied
Maintenance	#F59E0B	4	Under Maintenance

- **Card Style:** White, rounded 8px, shadow 0 1px 3px rgba(0,0,0,0.1), padding 20px
- **Icon:** 40px circle background (very light tint of icon color), icon centered
- **Value:** 28px Bold, #111827
- **Label:** 13px Regular, #6B7280

Ward Filter Tabs:

- Horizontal row of pill buttons: “All Wards” | “General” | “ICU” | “Emergency” | “Private”
- Active tab: Primary background, white text
- Inactive: #F3F4F6 background, #4B5563 text
- Margin-top: 24px from stats row

Bed Grid (Main Visual):

- **Layout:** Responsive grid, 4–6 columns depending on window width
- **Bed Card Size:** ~180×140px
- **Card Style:**
 - White background, rounded 8px, border 1px #E5E7EB
 - Top colored strip: 4px height — Green (Available), Red (Occupied), Yellow (Maintenance)
 - Padding: 16px
- **Bed Number:** 16px Bold, #111827, top-left
- **Ward Badge:** Pill badge top-right (e.g., “ICU”), background #F3F4F6, text #4B5563
- **Status Badge:** Below bed number, margin-top 8px (Available/Occupied/Maintenance pill)
- **Patient Name:** If occupied, 12px Regular, #4B5563, “Rahul Singh”
- **Days:** If occupied, 11px, #9CA3AF, “Day 3”

- **Equipment:** Small icons row at bottom (oxygen, ventilator, monitor)

Interactions:

- Click Available bed → Opens Admit Dialog pre-filled with that bed
- Click Occupied bed → Opens Discharge Dialog for that patient
- Click Maintenance bed → Shows maintenance details tooltip
- Right-click any bed → Context menu: “View Details”, “Mark Maintenance”, “History”
- Hover card → Slight lift shadow `0 4px 6px rgba(0,0,0,0.1)`

Recent Activity Panel (Bottom Right, optional):

- Card titled “Recent Activity”
- List of last 5 admissions/discharges with timestamp
- Small colored dot (green=admit, red=discharge)

3.3 Admit Patient Dialog

Type: Modal Dialog (JDialog)

Size: 600×650

Backdrop: Overlay `rgba(0,0,0,0.5)` on parent window

Animation: Fade in 150ms, slide up 20px

Header:

- Title: “Admit Patient”, 18px Bold
- Close button (X) top-right, #9CA3AF, hover #EF4444
- Bottom border 1px #E5E7EB

Body (Scrollable JPanel with padding 24px):

Section 1: Patient Information

- Section title: “Patient Information”, 14px Semibold, #111827
- Divider line below title
- **Two-column layout:**
 - Left: “Patient ID” label + input (if existing patient) OR “New Patient” toggle
 - Right: “Full Name” label + input
- **Two-column:**
 - Left: “Age” label + number input (spinner, 0-150)
 - Right: “Blood Group” label + dropdown (A+, A-, B+, B-, O+, O-, AB+, AB-)
- **Two-column:**
 - Left: “Contact” label + phone input
 - Right: “Emergency Contact” label + phone input

- **Full width:** “Address” label + multi-line text area (2 rows)

Section 2: Admission Details

- Section title: “Admission Details”, 14px Semibold
- **Two-column:**
- Left: “Ward Type” label + dropdown (General, ICU, Emergency, Private)
- Right: “Preferred Bed” label + dropdown (optional, lists available beds in selected ward)
- **Two-column:**
- Left: “Assigned Doctor” label + dropdown (populated from staff table, Doctors only)
- Right: “Severity Level” label + dropdown (1-Low to 5-Critical)
- **Two-column:**
- Left: “Admission Date” label + date picker (default today)
- Right: “Expected Discharge” label + date picker

Section 3: Notes

- “Admission Notes” label + text area (3 rows)

Footer (Sticky bottom, white background, top border 1px #E5E7EB):

- **Left:** “Cancel” button — Secondary style
- **Right:** “Admit Patient” button — Primary style, disabled until all required fields valid
- **Right of that:** “Add to Waitlist” button — Warning style (only enabled if no beds available in selected ward)

Validation:

- Name: Required, min 2 chars, letters only
- Age: Required, 0-150
- Contact: Required, 10 digits
- Ward Type: Required
- Doctor: Required
- Expected Discharge: Must be after Admission Date

Error Display:

- Invalid fields get red border #EF4444 and small red text below: “This field is required”
- Dialog shakes if submit attempted with errors

3.4 Discharge Dialog

Type: Modal Dialog

Size: 500×450

Trigger: Click occupied bed OR select from sidebar menu

Header: “Discharge Patient”, 18px Bold

Body:

- **Patient Card (highlighted):**
- Large text: Patient Name, 20px Bold
- Subtitle: “Bed ICU-03 · Admitted 3 days ago”
- Patient details row: Age, Blood Group, Contact
- Doctor: “Dr. Rajesh Sharma”
- **Discharge Details:**
 - “Actual Discharge Date” — Date picker (default today), disabled (auto-filled)
 - “Discharge Notes” — Text area (2 rows)
- **Bed Status After Discharge:**
 - Radio buttons:
 - * “Make Available Immediately” (default selected)
 - * “Send for Cleaning/Maintenance”
 - If Maintenance selected: “Reason” dropdown (Cleaning, Equipment Check, Repair, Other) + text field for Other

Footer:

- “Cancel” — Secondary
- “Confirm Discharge” — Danger style (red background)

Confirmation Step:

- On click “Confirm Discharge”, show confirmation overlay within dialog:
- “Are you sure you want to discharge [Patient Name]?”
- “Yes, Discharge” (Danger) | “No, Go Back” (Secondary)

Post-Discharge:

- If waitlist auto-allocation occurs, show success toast: “Bed auto-allocated to waitlisted patient: [Name]”
- Toast appears top-right of main window, auto-dismiss 4 seconds, green background

3.5 Waitlist Panel

Layout: Full main content area (when selected from sidebar)

Header Row:

- Title: “Waitlist Queue”, 20px Bold
- Right: “Total Waiting: 8” badge (warning color pill)
- Right: “Auto-Allocate All” button (Primary, only enabled if beds available)

Filter Tabs:

- “All” | “ICU” | “Emergency” | “General” | “Private”

- Same pill style as dashboard

Data Table:

- Columns: Priority Score | Patient Name | Age | Ward Requested | Waiting Time | Status | Actions
- **Priority Score:** Bold, color-coded — 500+ Red (critical), 300-500 Orange, <300 Blue
- **Waiting Time:** Auto-updating (e.g., “2h 15m”), refreshes every minute
- **Status:** Pill badge — “Waiting” (warning), “Allocated” (success)
- **Actions:**
- “Allocate” button (Primary, small) — only if matching bed available
- “Remove” button (Secondary, small) — removes from waitlist
- “View” icon — opens patient detail

Row Color Coding:

- Critical patients (score >400): Very light red background #FEF2F2
- Hover: #FF66FF

Empty State:

- Large checkmark icon, green
- “All caught up! No patients in waitlist.”
- Centered, generous padding

3.6 Reports Center

Layout: Full main content area

Header: “Reports & Analytics”, 20px Bold

Report Cards Grid (2 columns):

Report Card	Icon	Description
Occupancy Overview	Pie chart icon	Real-time bed occupancy by ward
Bed Turnover	Clock icon	Average stay duration per ward
Doctor Workload	User icon	Current admissions per doctor
Waitlist History	Clock icon	Historical waitlist trends
Maintenance Log	Wrench icon	Bed maintenance history
Admission Trends	Trending up icon	Daily admission/discharge counts

- **Card Style:** White, rounded 8px, padding 24px, hover shadow increase
- **Icon:** 48px, Primary color, top-left
- **Title:** 16px Semibold
- **Description:** 13px Regular, #6B7280
- **Click:** Opens detailed report view

Detailed Report View (Example: Occupancy Overview):

- Back button (arrow + “Reports”)
 - Title: “Occupancy Overview”
 - Date range picker: “Last 7 Days” | “Last 30 Days” | “Custom”
 - **Summary Cards (3):**
 - “Average Occupancy Rate: 78%”
 - “Most Occupied Ward: ICU (95%)”
 - “Total Admissions: 142”
 - **Data Table:** Ward | Total Beds | Occupied | Available | Occupancy % | Trend
 - **Export Button:** “Export to CSV” (Secondary, small)
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3.7 Admin Panel

Layout: Full main content area, tabbed interface

Tabs: Wards | Beds | Staff | Maintenance Log

Wards Tab:

- “Add Ward” button top-right (Primary)
- Table: Ward ID | Type | Floor | Capacity | Daily Charge | Beds Count | Actions (Edit, Delete)
- Edit opens inline or dialog

Beds Tab:

- Filter: Ward dropdown + Status dropdown + Search
- Table: Bed ID | Ward | Number | Equipment | Status | Current Patient | Actions
- Bulk actions: “Mark Selected as Maintenance”

Staff Tab:

- “Add Staff” button
- Table: ID | Name | Role | Department | Phone | Current Admissions | Actions
- Role filter dropdown

Maintenance Log Tab:

- Table: Log ID | Bed | Start Time | End Time | Duration | Reason | Resolved By
 - Filter by date range, bed, status
-

4. Interaction Patterns

4.1 Toast Notifications

- **Position:** Top-right corner, stacked vertically with 8px gap
- **Types:**

- **Success:** Green left border, checkmark icon, white background
- **Error:** Red left border, X icon, white background
- **Warning:** Yellow left border, alert icon, white background
- **Info:** Blue left border, info icon, white background
- **Behavior:** Auto-dismiss 4 seconds, manual close (X), slide in from right
- **Max:** 5 toasts visible, older ones fade

4.2 Loading States

- **Button Loading:** Spinner icon replaces text, button disabled, opacity 70%
- **Table Loading:** Skeleton rows (gray shimmer blocks) for 1–3 rows
- **Page Loading:** Centered spinner with “Loading...” text
- **Background Tasks:** Use `SwingWorker` to prevent UI freeze

4.3 Confirmation Patterns

- **Destructive Actions:** Two-step confirmation (e.g., discharge, delete)
- **Modal Backdrop:** Click outside modal to cancel (optional setting)
- **Keyboard:** Escape to cancel, Enter to confirm

4.4 Form Validation

- **Real-time:** Validate on blur (field loses focus)
- **Submit-time:** Validate all fields, scroll to first error
- **Visual:** Red border + red text below field
- **Success:** Green checkmark icon inside field (right side)

4.5 Search & Filter

- **Global Search:** Top bar search box, searches patients and beds
- **Table Filters:** Inline dropdown filters per column
- **Search Results:** Dropdown with highlighted matches, Enter to select first

5. Accessibility Considerations

- All interactive elements have focus indicators (blue outline 2px)
 - Tab order follows visual order (left-to-right, top-to-bottom)
 - Color is not the only indicator — icons + text labels for all statuses
 - Minimum touch target: 44×44px for buttons
 - High contrast mode support (darker text, stronger borders)
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6. Responsive Behavior

Since Swing is desktop-only, “responsive” means:

- **Window Resize:** Bed grid reflows (4 cols → 3 cols → 2 cols)
- **Minimum Size:** 900×600, below this show scrollbars

- **Maximized:** Stats cards expand, bed grid shows 6 columns
- **Sidebar:** Collapses to icon-only mode at window width < 1000px (hover to expand)

End of UI/UX Design Plan